Bass Model Analysis: Samsung Micro RGB TV

Albert Hakobyan

01-03-2026

Libraries

```
libs <- c('ggplot2', 'ggpubr', 'knitr', 'diffusion')
load_libraries <- function(libs) {
  new_libs <- libs[!(libs %in% installed.packages()[, "Package"])]
  if (length(new_libs) > 0) { install.packages(new_libs) }
  lapply(libs, library, character.only = TRUE)}
load_libraries(libs)
```

Defining Bass Model Functions

Following the lecture formulas:

$$f(t) = \frac{(p+q)^2}{p} \cdot \frac{e^{-(p+q)t}}{\left[1 + \frac{q}{p} e^{-(p+q)t}\right]^2}$$

$$F(t) = \frac{1 - e^{-(p+q)t}}{1 + \frac{q}{p} e^{-(p+q)t}}$$

```
bass.f <- function(t, p, q) {
  ((p + q)^2 / p) * exp(-(p + q) * t) / (1 + (q / p) * exp(-(p + q) * t))^2
  }

bass.F <- function(t, p, q) {
  (1 - exp(-(p + q) * t)) /(1 + (q / p) * exp(-(p + q) * t))
  }
```

Part 3 — Historical Data: LG OLED TV Shipments (2013–2025)

```
oled_data <- read.csv("./data/OLED_TV_Global_Annual_Shipments_2013_2025.csv")
names(oled_data) <- c("year", "sales", "cum_sales", "source")
oled_data$t <- 1:nrow(oled_data)

kable(oled_data[, c("year", "sales", "cum_sales", "t")],
      caption = "Global OLED TV Annual Shipments (Million Units)",
      col.names = c("Year", "Annual Shipments (M)", "Cumulative (M)", "t"))
```

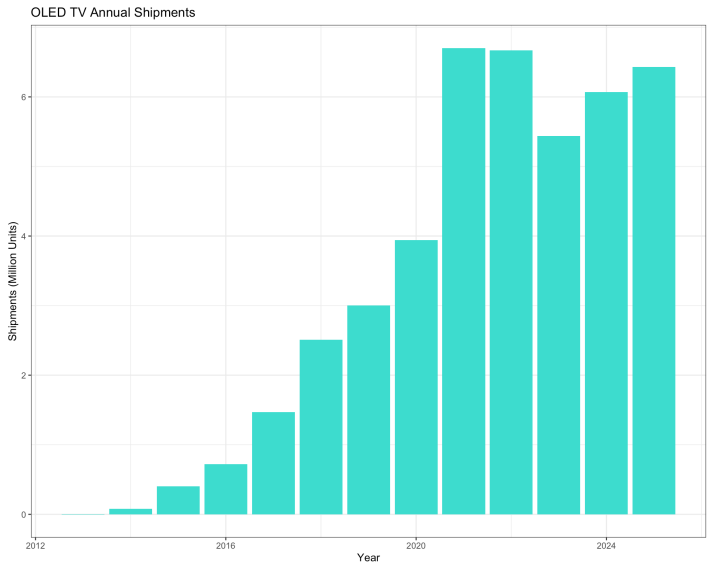
Global OLED TV Annual Shipments (Million Units)

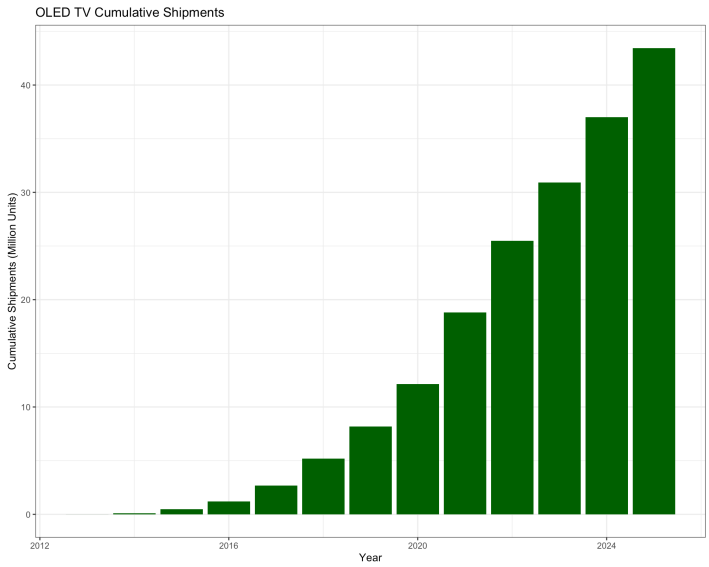
Year	Annual Shipments (M)	Cumulative (M)	t
2013	0.005	0.005	1
2014	0.077	0.082	2
2015	0.400	0.482	3
2016	0.720	1.202	4
2017	1.470	2.672	5
2018	2.510	5.182	6
2019	3.000	8.182	7
2020	3.940	12.122	8
2021	6.700	18.822	9
2022	6.670	25.492	10
2023	5.440	30.932	11
2024	6.070	37.002	12
2025	6.430	43.432	13

```
p1 <- ggplot(oled_data, aes(x = year, y = sales)) +
  geom_bar(stat = 'identity', fill = 'turquoise') +
  labs(title = 'OLED TV Annual Shipments',
       x = 'Year', y = 'Shipments (Million Units)') +
  theme_bw()

p2 <- ggplot(oled_data, aes(x = year, y = cum_sales)) +
  geom_bar(stat = 'identity', fill = 'darkgreen') +
  labs(title = 'OLED TV Cumulative Shipments',
       x = 'Year', y = 'Cumulative Shipments (Million Units)') +
  theme_bw()

ggarrange(p1, p2, ncol = 2)
```





> The 2022–2023 deviation from the fitted curve reflects macroeconomic shocks (European inflation, energy costs) and competitive pressure from MiniLED, which are exogenous factors not captured by the Bass diffusion framework.

Part 4 — Estimating Bass Model parameters

Method 1: Non-linear Least Squares (`nls`)

Using the closed-form Bass equation: $a(t) = m \cdot f(t)$

```
sales <- oled_data$sales
t <- 1:length(sales)

bass_m <- nls(sales ~ m * (((p + q)^2 / p) * exp(-(p + q) * t)) / (1 + (q / p) * e
xp(-(p + q) * t))^2,
              start = c(list(m = sum(sales), p = 0.02, q = 0.4)))

summary(bass_m)
```

```
##
## Formula: sales ~ m * (((p + q)^2/p) * exp(-(p + q) * t))/(1 + (q/p) *
##     exp(-(p + q) * t))^2
##
## Parameters:
##    Estimate Std. Error t value Pr(>|t|)
## m 59.718197   5.877942  10.160 1.37e-06 ***
## p  0.003160   0.001163   2.718   0.0216 *
## q  0.442787   0.053053   8.346 8.11e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6894 on 10 degrees of freedom
##
## Number of iterations to convergence: 11
## Achieved convergence tolerance: 7.208e-06
```

```
m_nls <- round(coef(bass_m)["m"], 4)
p_nls <- round(coef(bass_m)["p"], 4)
q_nls <- round(coef(bass_m)["q"], 4)

cat("NLS Estimates:\n")
```

```
## NLS Estimates:
```

```
cat("  m (market potential):", m_nls, "million units\n")
```

```
##   m (market potential): 59.7182 million units
```

```
cat("  p (innovation coeff):", p_nls, "\n")
```

```
##   p (innovation coeff): 0.0032
```

```
cat("  q (imitation coeff): ", q_nls, "\n")
```

```
##   q (imitation coeff):  0.4428
```

Method 2: Using the `diffusion` package's tools

The `diffusion` package estimates p, q, m as suggested by Bass.

```
diff_m <- diffusion(sales)
diff_m
```

```
## bass model
##
## Parameters:
##   Estimate p-value
## m  64.0267      NA
## p   0.0041      NA
## q   0.4085      NA
##
## sigma: 0.6566
```

```
m_diff <- round(diff_m$w, 4)[1]
p_diff <- round(diff_m$w, 4)[2]
q_diff <- round(diff_m$w, 4)[3]


cat("\nDiffusion Package Estimates:\n")
```

```
##
## Diffusion Package Estimates:
```

```
cat("  m (market potential):", m_diff, "million units\n")
```

```
##   m (market potential): 64.0267 million units
```

```
cat("  p (innovation coeff):", p_diff, "\n")
```

```
##   p (innovation coeff): 0.0041
```

```
cat("  q (imitation coeff): ", q_diff)
```

```
##   q (imitation coeff):  0.4085
```

Comparing both methods

```
comparison <- data.frame(
  Parameter = c("p (innovation)", "q (imitation)", "m (market potential, M unit
s)"),
  NLS = c(p_nls, q_nls, m_nls),
  Diffusion = c(p_diff, q_diff, m_diff)
)
kable(comparison, caption = "Parameter Estimates Comparison")
```

Parameter Estimates Comparison

Parameter	NLS	Diffusion

p	p (innovation)	0.0032	0.0041
q	q (imitation)	0.4428	0.4085
m	m (market potential, M units)	59.7182	64.0267

I'll will use the **NLS estimates** for the analysis (it seems to fit the nonlinear curve better).

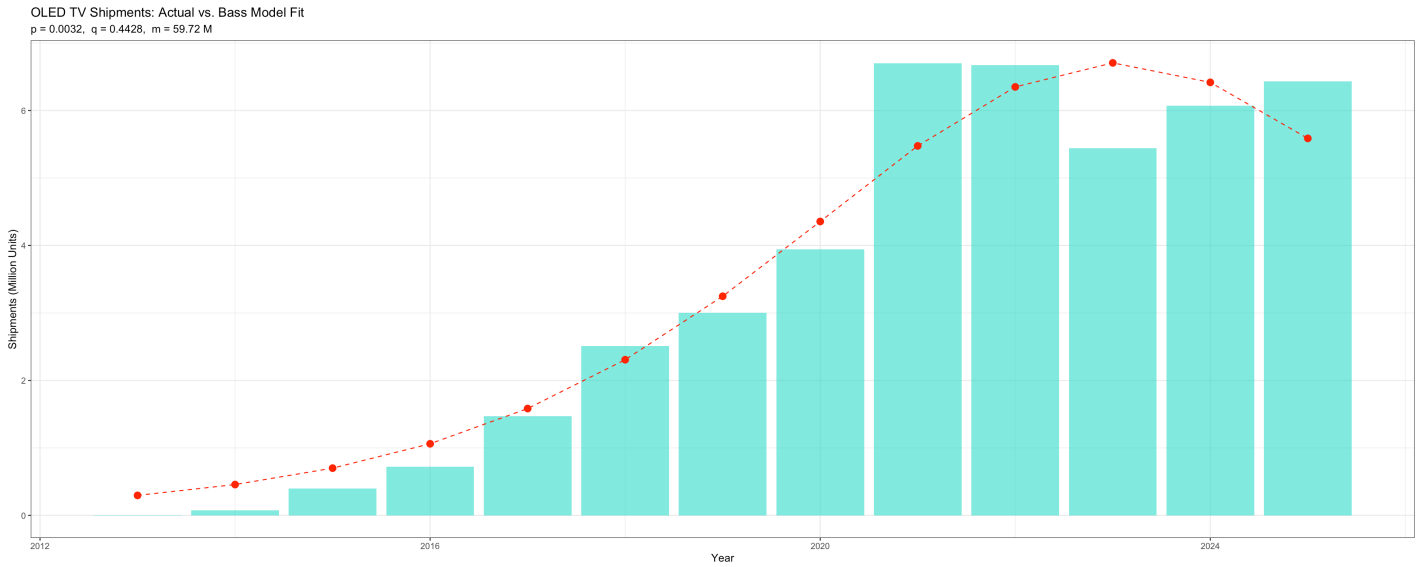
```
# Diffusion = c(p_diff, q_diff, m_diff)
#p_est <- as.numeric(0.0015) # this is my approximation after tweaking this parame
ter up and down > But I prefer the NLS estimate
#q_est <- as.numeric(0.5) # this is my approximation after tweaking this parameter
up and down > But I prefer the NLS estimate
#m_est <- as.numeric(60)
#
#p_est <- as.numeric(p_diff)
#q_est <- as.numeric(q_diff)
#m_est <- as.numeric(m_diff)
#
#


p_est <- as.numeric(p_nls)
q_est <- as.numeric(q_nls)
m_est <- as.numeric(m_nls)
```

Visualizing the Fitted Model vs. Actual Data

```
oled_data$pred_sales <- bass.f(1:nrow(oled_data), p = p_est, q = q_est) * m_est

ggplot(oled_data, aes(x = year, y = sales)) +
  geom_bar(stat = 'identity', fill = 'turquoise', alpha = 0.7) +
  geom_point(aes(y = pred_sales), color = 'red', size = 3) +
  geom_line(aes(y = pred_sales), color = 'red', linetype = 'dashed') +
  labs(title = 'OLED TV Shipments: Actual vs. Bass Model Fit',
       subtitle = paste0('p = ', p_est, ',  q = ', q_est, ',  m = ', round(m_est,
2), ' M'),
       x = 'Year', y = 'Shipments (Million Units)') +
  theme_bw()
```

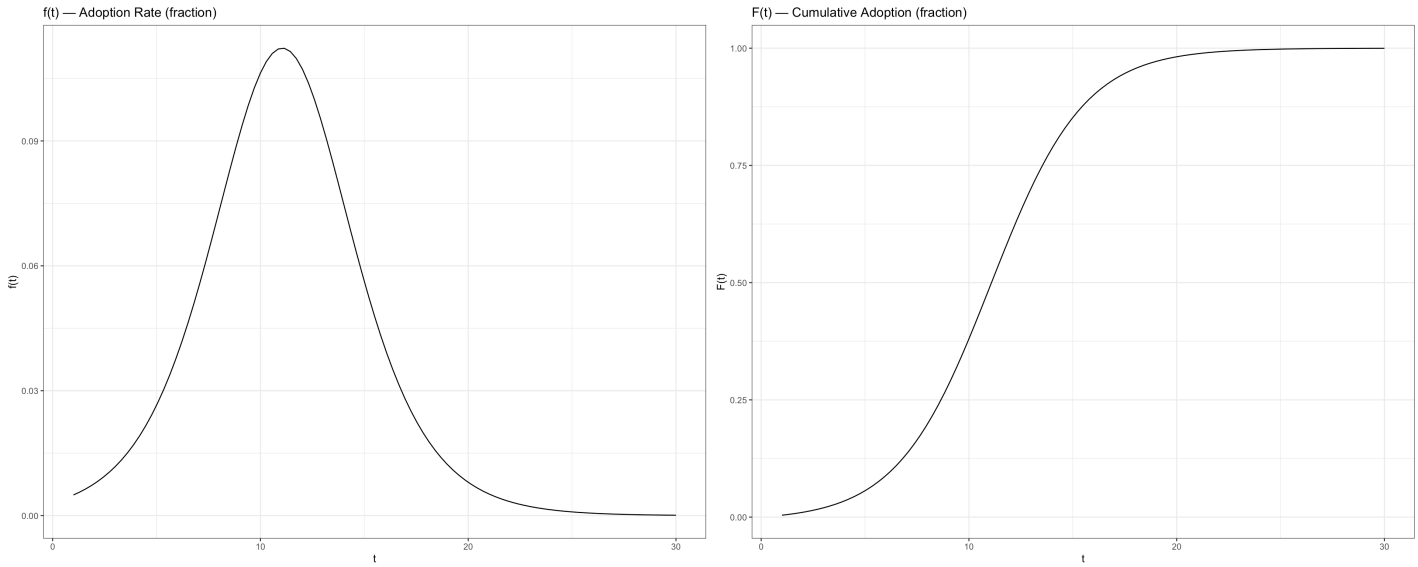


Plots of f(t) and F(t) theoretical curves using estimated parameters

```
# f(t) and F(t) theoretical curves using estimated parameters
ft_plot <- ggplot(data.frame(t = c(1, 30)), aes(t)) +
  stat_function(fun = bass.f, args = c(p = p_est, q = q_est)) +
  labs(title = 'f(t) — Adoption Rate (fraction)', x = 't', y = 'f(t)') +
  theme_bw()

Ft_plot <- ggplot(data.frame(t = c(1, 30)), aes(t)) +
  stat_function(fun = bass.F, args = c(p = p_est, q = q_est)) +
  labs(title = 'F(t) — Cumulative Adoption (fraction)', x = 't', y = 'F(t)') +
  theme_bw()

ggarrange(ft_plot, Ft_plot, ncol = 2)
```



Time to Peak Sales

$$t^* = \frac{\ln(q/p)}{p+q}$$

```
t_star <- log(q_est / p_est) / (p_est + q_est)
peak_year_oled <- 2013 + t_star - 1  #t=1 corresponds to 2013

cat("Predicted peak period (t*):", round(t_star, 2), "\n")
```

```
## Predicted peak period (t*): 11.05
```

```
cat("Corresponding calendar year:", round(peak_year_oled, 1), "\n")
```

```
## Corresponding calendar year: 2023.1
```

```
cat("Actual peak year in data:  ", oled_data$year[which.max(oled_data$sales)], "\
n")
```

```
## Actual peak year in data:   2021
```

Task 5, 6, 7 — Predict the Diffusion of Samsung Micro RGB TV

Scope: Global Diffusion

I analyze diffusion worldwide because the look-alike product data (LG OLED TV) is global shipment data. Samsung also launches Micro RGB as a global product, so applying global p and q is consistent with that.

Estimating Market Potential (m) for Samsung Micro RGB TV

The LG OLED TV data gives us estimated p and q. For Samsung Micro RGB TV, we need our own estimate of **m** (market potential). Here is the Fermi estimation (same estimation is also done within the other report PDF file in the ./report directory):

```
# --- Fermi Estimation for Samsung Micro RGB TV Market Potential ---
#
# The global TV market ships about 200-230M units/year (TrendForce 2023; Counterpo
int 2025).
# Of those, roughly 8-10% fall in the premium segment above $1,500 — around 18M un
its/year.
# Our NLS fit estimates OLED TV's lifetime market potential at ~60M cumulative uni
ts.
#
# Samsung Micro RGB TV targets the same premium segment, but must probably lead to
steeper upward trend:
# it launched at $29,999 (about 2x the first LG OLED's $15,000 in 2013), Samsung i
s the
# sole manufacturer (vs. 20+ OLED TV brands by 2022), and it enters a market where
OLED
# and MiniLED are already mature competitors and the customers' perceived benefit
is not going to be the same as for the OLEDs (my oppinion).
#
# So I scale OLED's m down by 0.65 to account for all of the above-mentioned; it i
s a number which adds my personal bias.
# Samsung is an established brand for the ultra-premium segment so this coefficien
t may fluctuate as well: 0.60-0.75
#
# Sources:
#   TrendForce (Nov 2023) — ~197M TV shipments: https://www.trendforce.com/pressce
nter/news/20231128-11931.html
#   Counterpoint/Omdia (2025) — 230M units in 2024: https://displaydaily.com/tv-ma
rket-recovering-getting-bigger-brighter-better/
#   LG OLED launch price ($15K, 2013): https://www.oled-info.com/lgs-oled-tv-will-
arrive-uk-july-2013-9999-15000
#   20+ OLED TV brands by 2022: https://www.lg.com/global/newsroom/lg-story/beyond
-news/lg-oled-tv-a-decade-at-the-top-and-no-sign-of-slowing-down/

m_micro_rgb <- m_est * 0.65
cat("Estimated m for Samsung Micro RGB TV:", round(m_micro_rgb, 1), "million units
\n")
```

```
## Estimated m for Samsung Micro RGB TV: 38.8 million units
```

```
cat("(Based on OLED TV look-alike diffusion potential, scaled by 0.65)\n")
```

```
## (Based on OLED TV look-alike diffusion potential, scaled by 0.65)
```

Predicted Diffusion Curve for Samsung Micro RGB TV

Samsung Micro RGB TV launched in **2025** (t = 1).

```
# Prediction horizon: 2025 to 2040 (15 years)
forecast_years <- 2025:2040
t_forecast <- 1:length(forecast_years)

# Predicted annual adoptions
pred_annual <- bass.f(t_forecast, p = p_est, q = q_est) * m_micro_rgb

# Predicted cumulative adoptions
pred_cumulative <- bass.F(t_forecast, p = p_est, q = q_est) * m_micro_rgb

micro_rgb_forecast <- data.frame(year = forecast_years,
                                 t = t_forecast,
                                 predicted_annual = round(pred_annual, 4),
                                 predicted_cumulative = round(pred_cumulative, 4))

kable(micro_rgb_forecast,
      caption = "Predicted Samsung Micro RGB TV Diffusion (Million Units)",
      col.names = c("Year", "t", "Annual Adoptions (M)", "Cumulative Adoptions (
M)"))
```

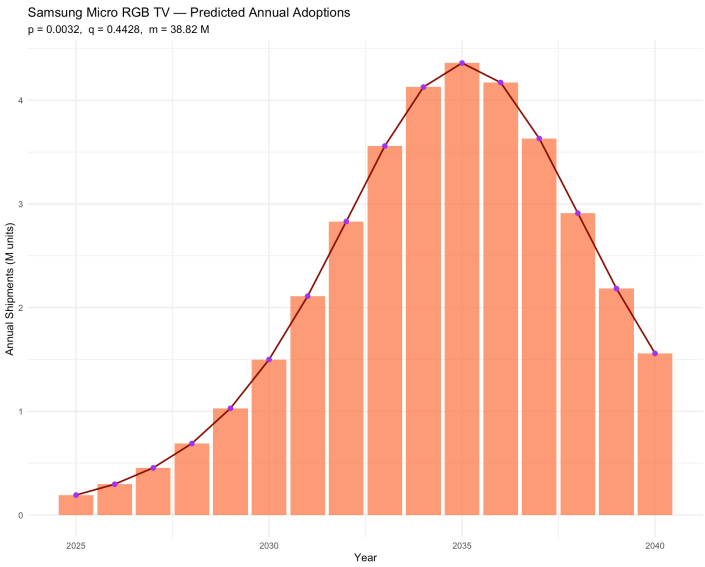
Predicted Samsung Micro RGB TV Diffusion (Million Units)

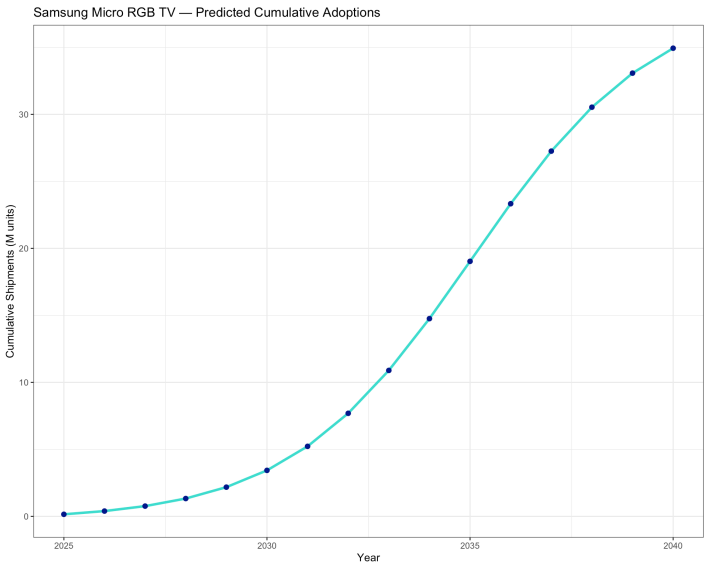
Year	t	Annual Adoptions (M)	Cumulative Adoptions (M)
2025	1	0.1925	0.1559
2026	2	0.2969	0.3969
2027	3	0.4549	0.7675
2028	4	0.6896	1.3323
2029	5	1.0290	2.1816
2030	6	1.4993	3.4340
2031	7	2.1105	5.2277
2032	8	2.8307	7.6922
2033	9	3.5591	10.8927
2034	10	4.1272	14.7573
2035	11	4.3587	19.0339
2036	12	4.1709	23.3331
2037	13	3.6313	27.2572
2038	14	2.9111	30.5355
2039	15	2.1839	33.0779
2040	16	1.5583	34.9380

```
p_annual <- ggplot(micro_rgb_forecast, aes(x = year, y = predicted_annual)) +
  geom_bar(stat = 'identity', fill = 'coral', alpha = 0.8) +
  geom_line(color = 'darkred', linewidth = 0.8) +
  geom_point(color = 'purple', size = 2) +
  labs(title = 'Samsung Micro RGB TV — Predicted Annual Adoptions',
       subtitle = paste0('p = ', p_est, ',  q = ', q_est, ',  m = ', round(m_micro
_rgb, 2), ' M'),
       x = 'Year', y = 'Annual Shipments (M units)') +
  theme_minimal()

p_cum <- ggplot(micro_rgb_forecast, aes(x = year, y = predicted_cumulative)) +
  geom_line(color = 'turquoise', linewidth = 1.2) +
  geom_point(color = 'darkblue', size = 2) +
  labs(title = 'Samsung Micro RGB TV — Predicted Cumulative Adoptions',
       x = 'Year', y = 'Cumulative Shipments (M units)') +
  theme_bw()

ggarrange(p_annual, p_cum, ncol = 2)
```





Peak sales timing for Samsung Micro RGB TV

```
t_star_micro <- log(q_est / p_est) / (p_est + q_est)
peak_year_micro <- 2025 + t_star_micro

cat("Predicted peak period (t*):", round(t_star_micro, 2), "years after launch\n")
```

```
## Predicted peak period (t*): 11.05 years after launch
```

```
cat("Predicted peak calendar year:", round(peak_year_micro, 1), "\n")
```

```
## Predicted peak calendar year: 2036.1
```

```
cat("Predicted peak annual shipments:",
    round(bass.f(t_star_micro, p = p_est, q = q_est) * m_micro_rgb, 2), "million u
nits")
```

```
## Predicted peak annual shipments: 4.36 million units
```

Part 7 — Estimating the Number of Adopters by Period

```
# Computing innovators vs imitators at each time period
# From the Bass equation:
#   Total adopters at t:      a(t) = m * f(t)
#   Innovator component:      p * [m - A(t-1)]
#   Imitator component:       q * [A(t-1)/m] * [m - A(t-1)]

adopters <- data.frame(year = forecast_years,
                       t = t_forecast,
                       annual_total = pred_annual,
                       cumulative = pred_cumulative)

# Computing innovator and imitator portions
adopters$remaining <- m_micro_rgb - c(0, adopters$cumulative[-nrow(adopters)])
adopters$cum_prev  <- c(0, adopters$cumulative[-nrow(adopters)])

adopters$innovators <- p_est * adopters$remaining
adopters$imitators  <- q_est * (adopters$cum_prev / m_micro_rgb) * adopters$remain
ing

kable(adopters[, c("year", "annual_total", "innovators", "imitators", "cumulativ
e")],
      digits = 4,
      caption = "Adopters by Period: Innovators vs. Imitators (Million Units)",
      col.names = c("Year", "Total Adopters (M)", "Innovators (M)",
                    "Imitators (M)", "Cumulative (M)"))
```

Adopters by Period: Innovators vs. Imitators (Million Units)

Year	Total Adopters (M)	Innovators (M)	Imitators (M)	Cumulative (M)
2025	0.1925	0.1242	0.0000	0.1559
2026	0.2969	0.1237	0.0688	0.3969
2027	0.4549	0.1229	0.1740	0.7675
2028	0.6896	0.1218	0.3331	1.3323
2029	1.0290	0.1200	0.5697	2.1816

2030	1.4993	0.1172	0.9117	3.4340
2031	2.1105	0.1132	1.3861	5.2277
2032	2.8307	0.1075	2.0031	7.6922
2033	3.5591	0.0996	2.7311	10.8927
2034	4.1272	0.0894	3.4698	14.7573
2035	4.3587	0.0770	4.0502	19.0339
2036	4.1709	0.0633	4.2954	23.3331
2037	3.6313	0.0495	4.1213	27.2572
2038	2.9111	0.0370	3.5943	30.5355
2039	2.1839	0.0265	2.8846	33.0779
2040	1.5583	0.0184	2.1655	34.9380

Summary

```
summary_df <- data.frame(
  Item = c("Innovation (p)", "Imitation (q)",
           "Market Potential (m, M units)",
           "Predicted Peak Year (Samsung Micro RGB)",
           "Peak Annual Shipments (M units)",
           "Scope"),
  Value = c(p_est, q_est,
            round(m_micro_rgb, 2),
            round(peak_year_micro, 1),
            round(bass.f(t_star_micro, p = p_est, q = q_est) * m_micro_rgb, 2),
            "Global")
)
kable(summary_df, caption = "Bass Model Summary for Samsung Micro RGB TV")
```

Bass Model Summary for Samsung Micro RGB TV

Item	Value
Innovation (p)	0.0032
Imitation (q)	0.4428
Market Potential (m, M units)	38.82
Predicted Peak Year (Samsung Micro RGB)	2036.1
Peak Annual Shipments (M units)	4.36
Scope	Global